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Koonath

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(54) **SEPARATION OF LIGHT SIGNALS IN A LIDAR SYSTEM**

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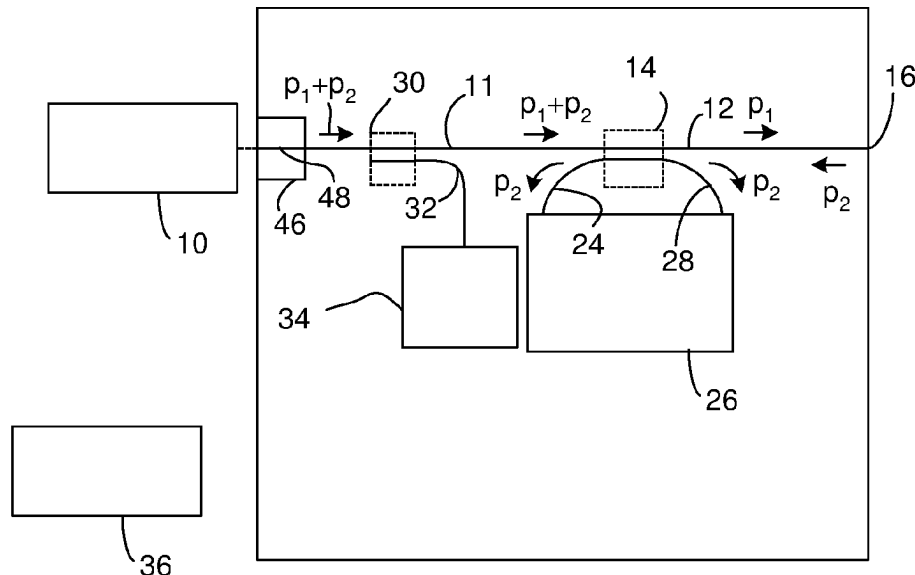
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(57) **ABSTRACT**

A LIDAR system includes a LIDAR chip with a utility waveguide configured to guide an outgoing LIDAR signal and an incoming LIDAR signal. The incoming LIDAR signal includes light from the LIDAR output signal after an object located outside of the LIDAR system reflects the light from the LIDAR output signal. The LIDAR chip also includes a polarizing-beam splitter configured to receive the outgoing LIDAR signal and the incoming LIDAR signal and to separate the incoming LIDAR signal from the outgoing LIDAR signal.

18 Claims, 8 Drawing Sheets



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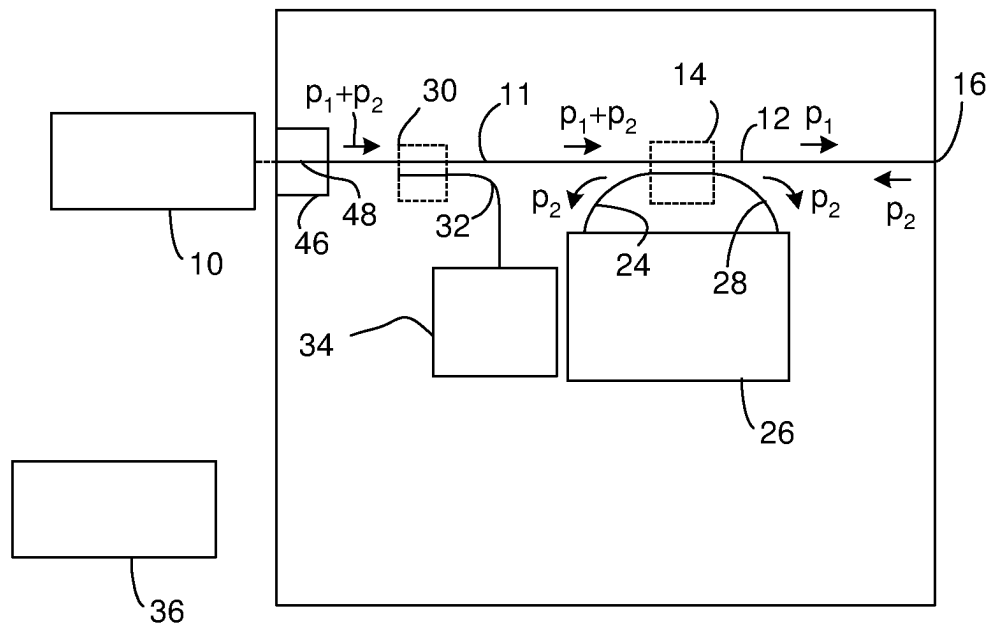


Figure 1

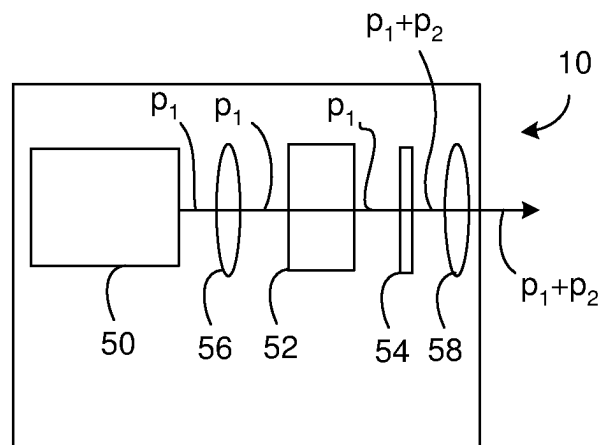
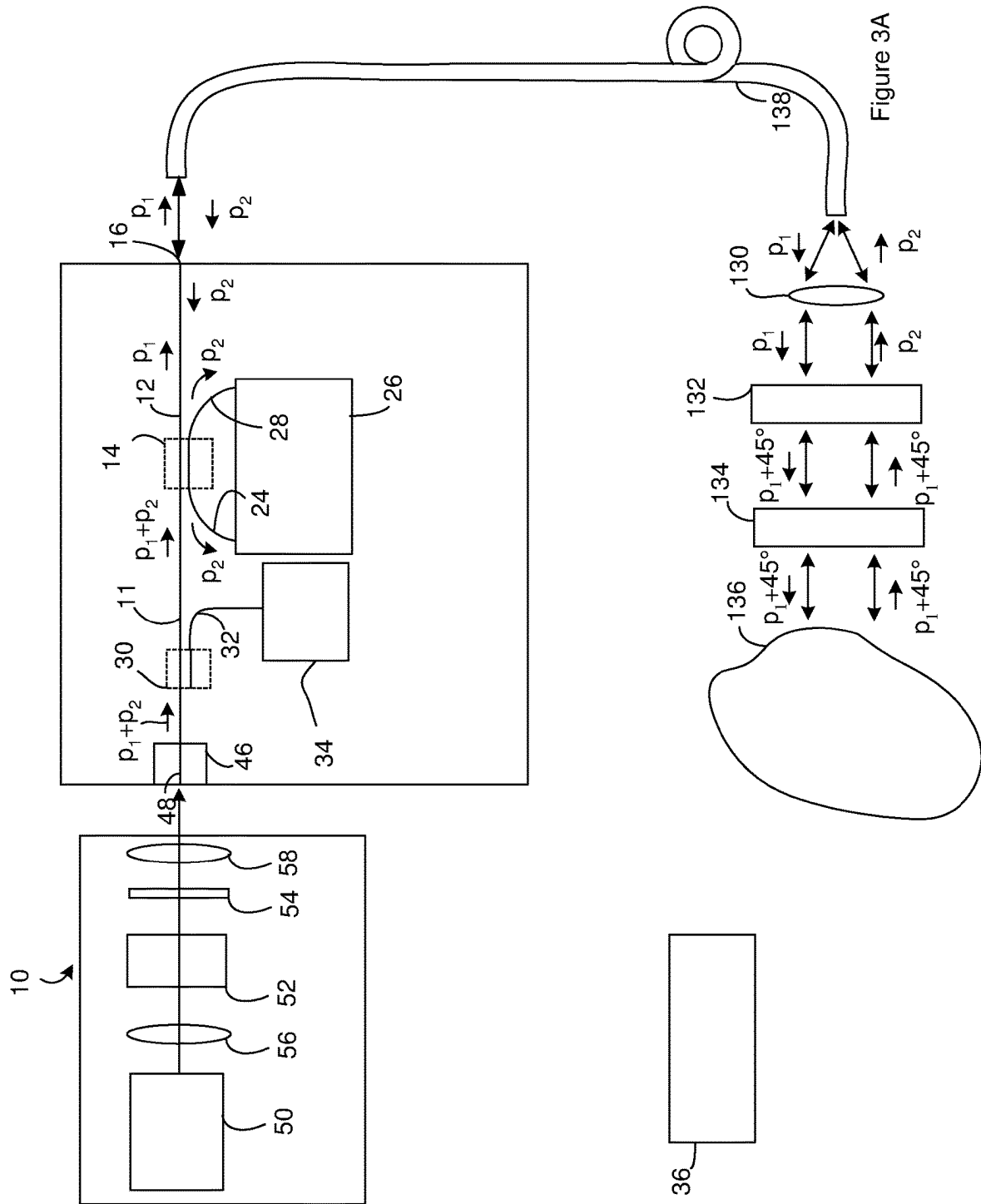


Figure 2



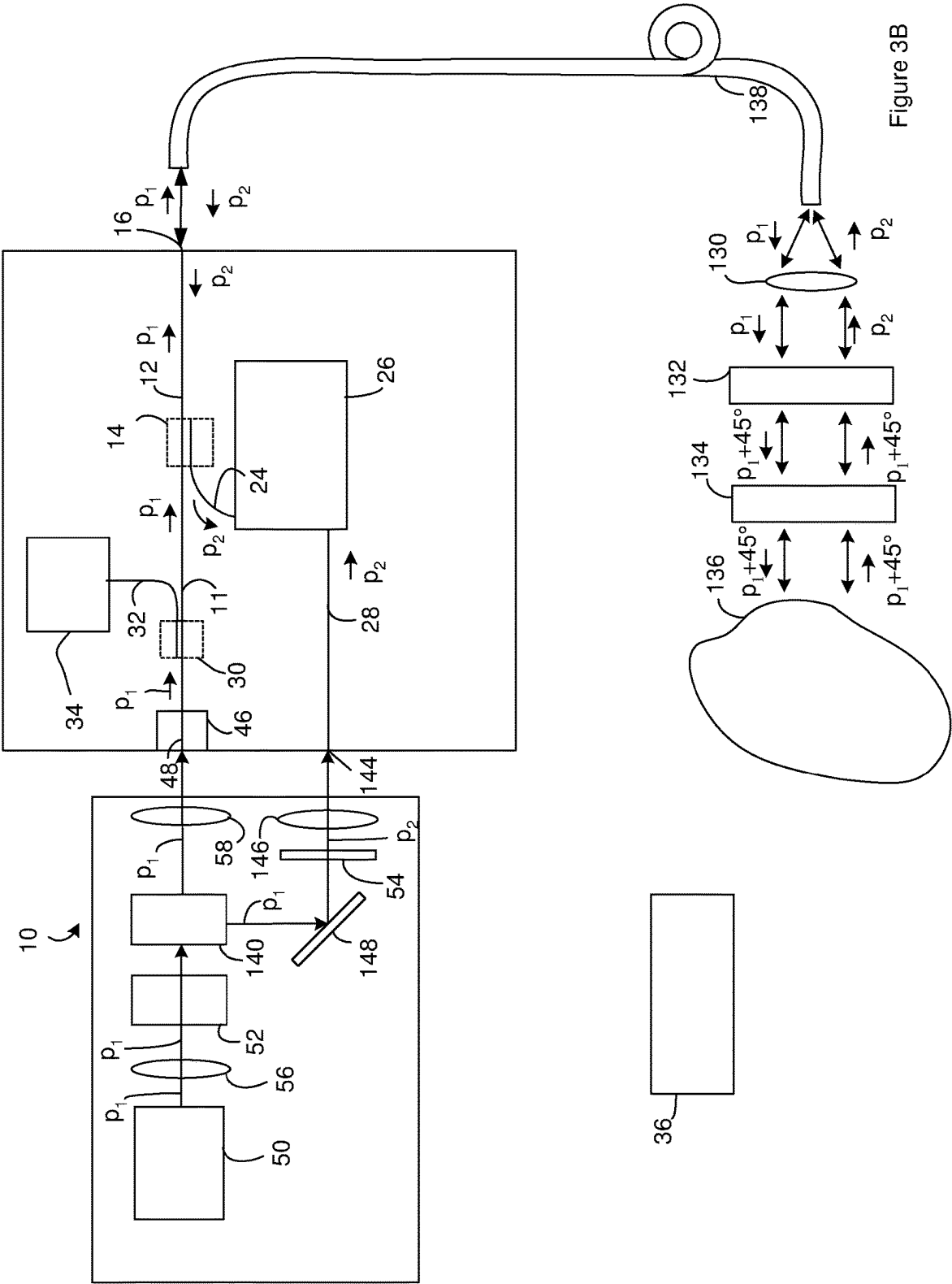


Figure 3B

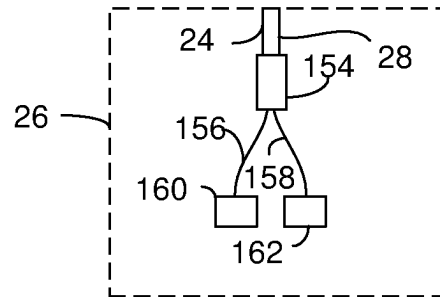


Figure 4A

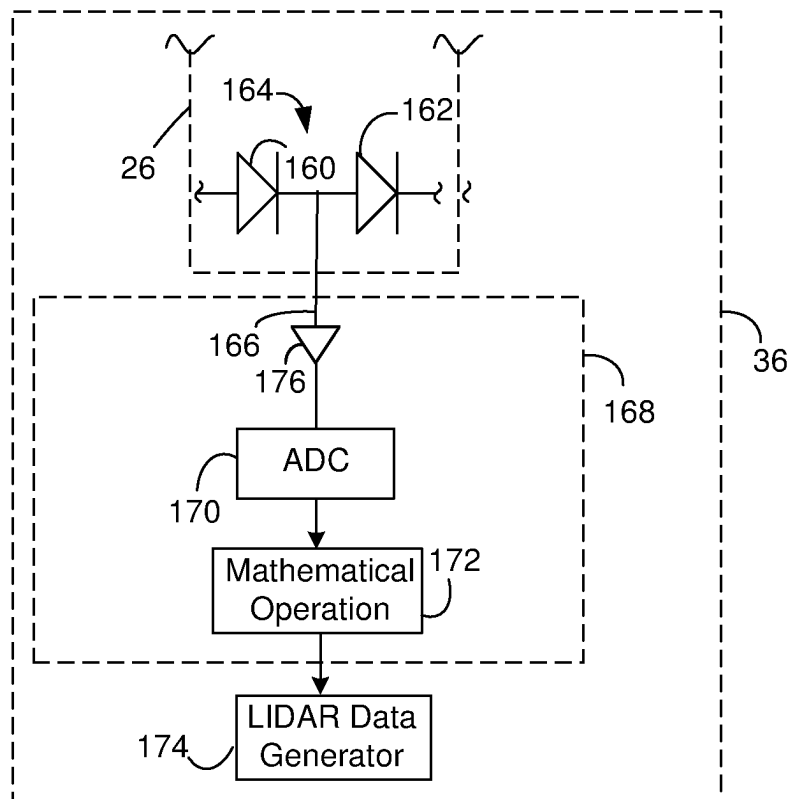


Figure 4B

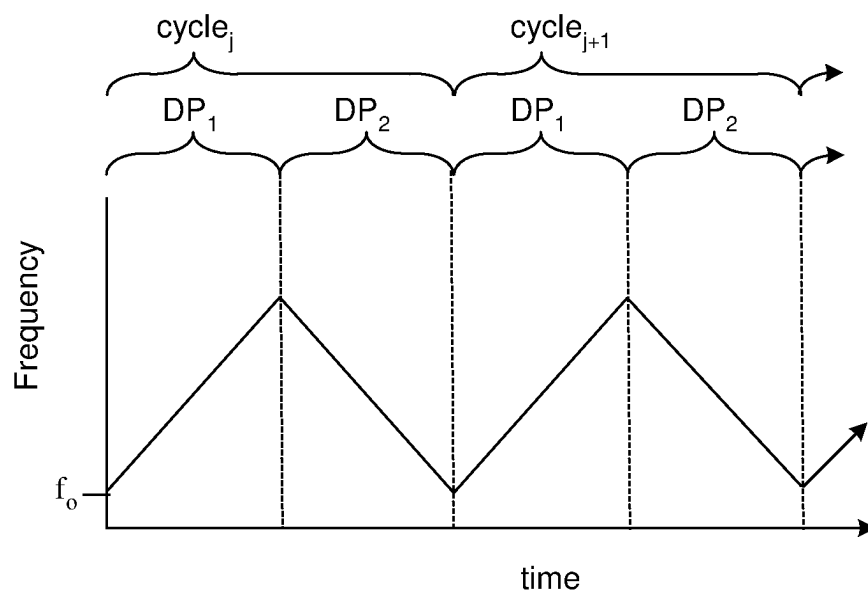


Figure 4C

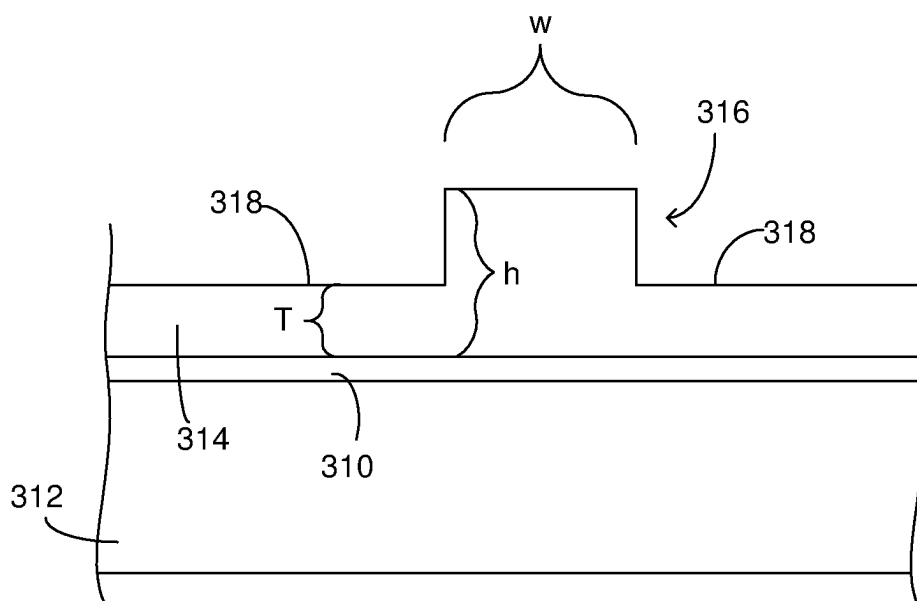


Figure 5

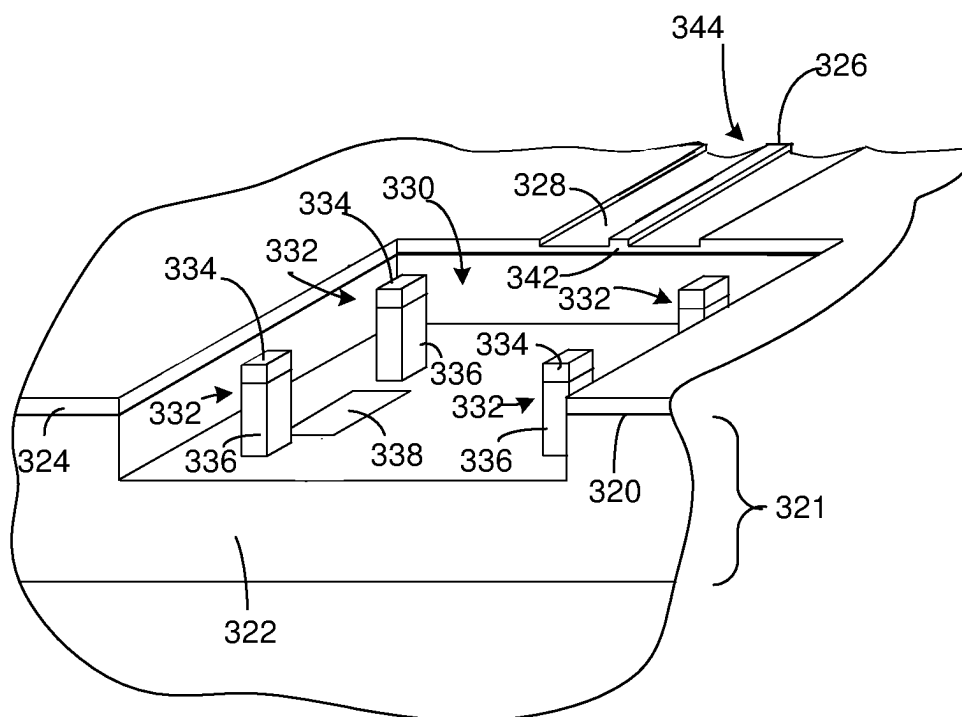


Figure 6A

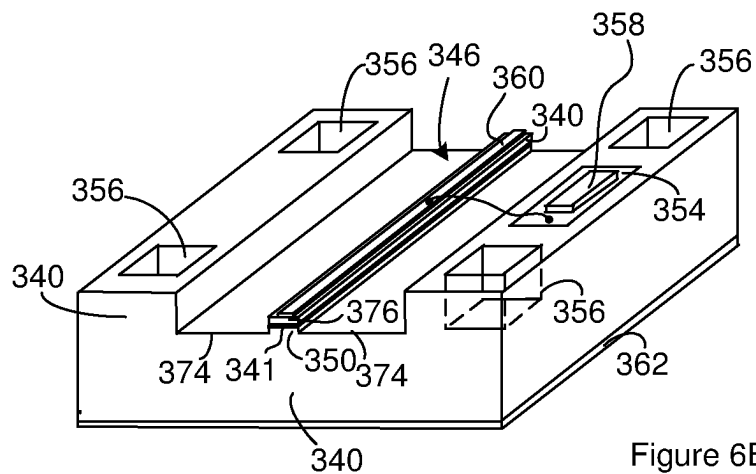


Figure 6B

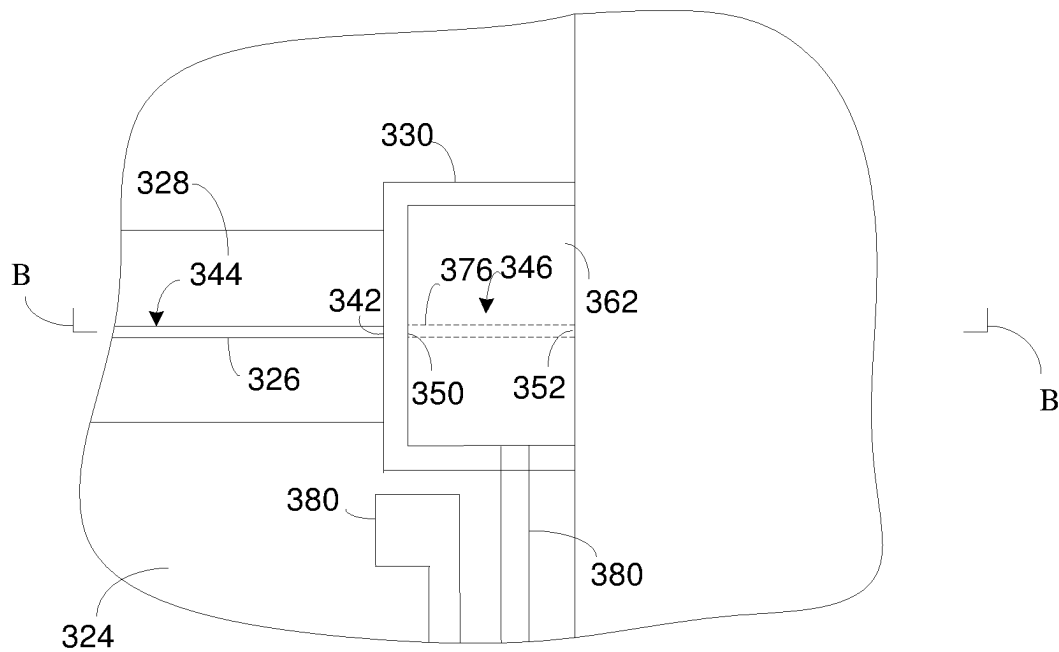


Figure 6C

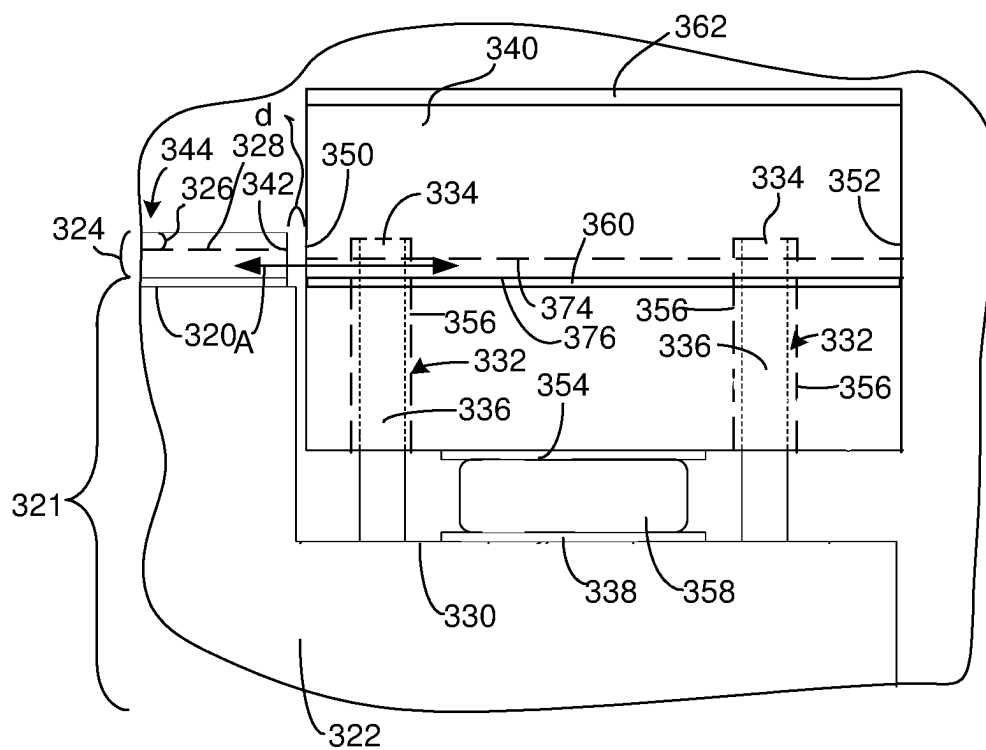


Figure 6D

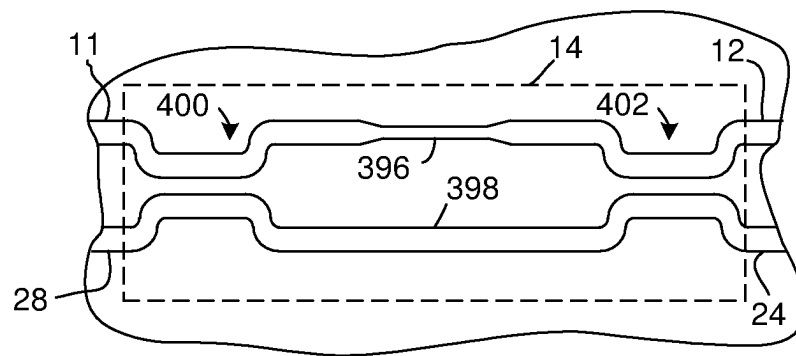


Figure 7

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SEPARATION OF LIGHT SIGNALS IN A LIDAR SYSTEM

FIELD

The invention relates to optical devices. In particular, the invention relates to LIDAR systems.

BACKGROUND

LIDAR systems bounce light off objects located outside of the LIDAR system. The LIDAR system uses light that returns to the LIDAR system from the object to calculate LIDAR data that indicates the distance and/or radial velocity between the LIDAR system and the object.

LIDAR systems can often be described as coaxial or biaxial. In coaxial systems, the outgoing light that is to be transmitted from the LIDAR system and the light that returns to the LIDAR system often travel in opposing directions along the same waveguide. However, the returning light needs to be separated from the outgoing light in order to process the returning light. This separation has been achieved with an optical coupler such as a 3 dB coupler. However, the returning light often is at very low power levels because the returning light has been reflected by a variety of different objects that are located outside of the LIDAR system. Because of the low power levels of the returning light, the optical loss associated with the use of optical couplers reduces the reliability of LIDAR data generated from these LIDAR systems. As a result, there is a need for LIDAR systems that produce LIDAR data with increased levels of reliability.

SUMMARY

A LIDAR system includes a LIDAR chip with a utility waveguide configured to guide an outgoing LIDAR signal and an incoming LIDAR signal. The incoming LIDAR signal includes light from the LIDAR output signal after an object located outside of the LIDAR system reflects the light from the LIDAR output signal. The LIDAR chip also includes a polarizing-beam splitter configured to receive the outgoing LIDAR signal and the incoming LIDAR signal and to separate the incoming LIDAR signal from the outgoing LIDAR signal.

Another embodiment of a LIDAR system includes a LIDAR chip with a utility waveguide configured to guide an outgoing LIDAR signal and an incoming LIDAR signal. The incoming LIDAR signal includes light from the LIDAR output signal after an object located outside of the LIDAR system reflects the light from the LIDAR output signal. The LIDAR chip also includes a polarizing-beam splitter configured to couple a portion of the outgoing LIDAR signal from the utility waveguide onto a reference waveguide as a reference signal and to couple a portion of the incoming LIDAR signal from the utility waveguide onto a comparative waveguide as a comparative signal. The LIDAR chip generates a composite signal having light from the reference signal beating with light from the comparative signal. The LIDAR system also includes electronics configured to use the beat frequency of the composite signal as a variable in a calculation of LIDAR data that indicates a distance and/or radial velocity between the LIDAR system and the object.

BRIEF DESCRIPTION OF THE FIGURES

FIG. 1 is a topview of a schematic of a LIDAR chip for use in a LIDAR system.

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FIG. 2 is a topview of a schematic of a light source that is suitable for use with the LIDAR chip of FIG. 1.

FIG. 3A is a topview of a LIDAR system that includes the LIDAR chip of FIG. 1 and the LIDAR source of FIG. 2.

FIG. 3B is a topview of a LIDAR system.

FIG. 4A is a schematic of an example of a suitable processing component for use in a LIDAR system.

FIG. 4B is a schematic of the relationship between the electronics and the light sensors in a processing.

FIG. 4C shows an example of a suitable frequency pattern for the system output signal.

FIG. 5 is a cross-section of a portion of a LIDAR chip that includes a waveguide on a silicon-on-insulator platform.

FIG. 6A is a perspective view of a portion of a LIDAR chip that includes an interface for optically coupling the LIDAR chip with an amplifier.

FIG. 6B is a perspective view of an amplifier chip suitable for use with the portion of the LIDAR chip shown in FIG. 6A.

FIG. 6C and FIG. 6D illustrate system that includes the LIDAR chip of FIG. 6A interfaced with the amplifier of FIG. 6B. FIG. 6C is a topview of the system.

FIG. 6D is a cross section of the system shown in FIG. 6C taken through a waveguide on the LIDAR chip and the amplifier waveguide on the amplifier chip.

FIG. 7 is a topview of a polarizing beam-splitter suitable for use with the LIDAR chip of FIG. 1.

DESCRIPTION

A LIDAR system includes a polarizing-beam splitter that receives an input light signal from an input waveguide and outputs an outgoing LIDAR signal on a utility waveguide. The outgoing LIDAR signal includes or consists of light from the input signal. The LIDAR system transmits a system output signal that includes or consists of light from the outgoing LIDAR signal. The system output signal can be reflected by an object located outside of the LIDAR system. The reflected portion of the system output signal can serve as a system return signal. The LIDAR system can receive light from the system return signal. Light from the system return signal can be included in an incoming LIDAR signal that the polarizing-beam splitter receives from the utility waveguide. As a result, the utility waveguide carries both the incoming LIDAR signal and the outgoing LIDAR signal in opposing directions. The LIDAR system is configured such that the input light signal is primarily in a first polarization state and the incoming LIDAR signal is primarily in a second polarization state. The first polarization state is different from the second polarization state. The polarizing-beam splitter uses the difference in the polarization states of the input light signal and the incoming LIDAR signal to separate the incoming LIDAR signal from the input signal. The polarizing-beam splitter can output at least a portion of the incoming LIDAR signal on a comparative waveguide. Since the polarization state of the incoming LIDAR signal is used to separate the incoming LIDAR signal from other light signals, the optical loss associated with separation of the incoming LIDAR signal is reduced relative to systems that use a coupler to separate optical signals. Accordingly, the LIDAR system retains more of the incoming LIDAR signal than is retained by prior systems. The increase in the power level retention of the incoming LIDAR signal allows the LIDAR system to generate more reliable LIDAR data.

FIG. 1 is a topview of a schematic of a LIDAR chip that can serve as a LIDAR system or can be included in a LIDAR system that includes components in addition to the LIDAR

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chip. The LIDAR chip can include a Photonic Integrated Circuit (PIC) and can be a Photonic Integrated Circuit chip.

The LIDAR chip includes an input waveguide **11** that receives a light source signal from a light source **10**. The light source **10** is configured such that a first portion of the light source signal has a first polarization state (labeled p_1 in FIG. 1) and a second portion of the light source signal has a second polarization state (labeled p_2 in FIG. 1). The portion of the light source signal that enters the input waveguide **11** serves as an input signal. As a result, a first portion of the input signal has a first polarization state (labeled p_1 in FIG. 1) and a second portion of the input source signal has a second polarization state (labeled p_2 in FIG. 1).

The first polarization state and the second polarization state can be linear polarization states and the second polarization state is different from the first polarization state. For instance, the first polarization state can be TE and the second polarization state can be TM or the first polarization state can be TM and the second polarization state can be TE.

The input waveguide **11** carries the input signal to a splitter **14** that receives the input signal. The splitter can be a polarizing beam-splitter. When a polarizing beam-splitter receives an input signal that has light in multiple different polarization states, the polarizing beam-splitter splits the input signal into two different signals that each carries light of a different polarization state. The polarizing beam-splitter directs the different signals to different waveguides that carry the output of the polarizing beam-splitter. As a result, a polarizing beam-splitter directs light to one of multiple different waveguides in response to the polarization state of the light. Accordingly, the splitter **14** outputs on a utility waveguide **12** an outgoing LIDAR signal that includes primarily, consists essentially of, and/or consists of light from the first portion of the input signal. As a result, the light in the outgoing LIDAR signal can include primarily, consist essentially of, and/or consist of light in the first polarization state and, in some instances, can exclude or substantially exclude light in the second polarization state. A suitable polarizing beam-splitter includes, but is not limited to, a polarization splitter such as a 2x2 polarizing beam-splitter.

The utility waveguide **12** terminates at a facet **16** and carries the outgoing LIDAR signal to the facet **16**. The facet **16** can be positioned such that the outgoing LIDAR signal traveling through the facet **16** exits the LIDAR chip and serves as a LIDAR output signal. For instance, the facet **16** can be positioned at an edge of the chip so the outgoing LIDAR signal traveling through the facet **16** exits the chip and serves as the LIDAR output signal. In some instances, the portion of the LIDAR output signal that has exited from the LIDAR chip can also be considered a system output signal. As an example, when the exit of the LIDAR output signal from the LIDAR chip is also an exit of the LIDAR output signal from the LIDAR system, the LIDAR output signal can also be considered a system output signal.

Light from the LIDAR output signal travels away from the LIDAR system in the system output signal. The system output signal can travel through free space in the atmosphere in which the LIDAR system is positioned. The system output signal may be reflected by one or more objects in the path of the system output signal. When the system output signal is reflected, at least a portion of the reflected light travels back toward the LIDAR chip as a system return signal.

Light from the system return signal can be carried in a LIDAR input signal that is received by the LIDAR chip. In some instances, a portion of the system return signal can

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serve as a LIDAR input signal. The LIDAR input signal enters the utility waveguide **12** through the facet **16** and serves as an incoming LIDAR signal. The utility waveguide **12** carries the incoming LIDAR signal to the splitter **14**.

The splitter **14** outputs at least a portion of the incoming LIDAR signal on a comparative waveguide **24** as a comparative signal. As will be described below, the LIDAR input signal includes primarily, consists essentially of, and/or consists of light in the second polarization state and in some instances, can exclude or substantially exclude light in the first polarization state. As a result, the incoming LIDAR signal and the comparative signal can include primarily, consist essentially of, and/or consist of light in the second polarization state and in some instances, can exclude or substantially exclude light in the first polarization state.

The comparative waveguide **24** carries the incoming LIDAR signal to a processing component **26** for further processing. Additionally, the splitter **14** outputs on a reference waveguide **28** a reference signal that includes primarily, consists essentially of, and/or consists of light from the second portion of the input signal. As a result, the light in the reference signal can include primarily, consist essentially of, and/or consist of light in the second polarization state and, in some instances, can exclude or substantially exclude light in the first polarization state. In contrast with the comparative signal, the reference signal includes primarily, consists of, and/or consists essentially of light that has not exited from the LIDAR chip. Since the reference signal and the comparative signal both include light from the input signal, the input signal can serve as a common source signal for the reference signal and the comparative signal.

The reference waveguide **28** carries the reference signal to the processing component **26** for further processing. As a result, the LIDAR system is configured such that the reference signal and the comparative signal received by the processing component **26** each includes primarily, consists essentially of, and/or consists of light in the same polarization state.

The LIDAR chip can include a control branch for controlling operation of the light source **10**. The control branch includes a splitter **30** that moves a portion of the light source signal from the input waveguide **11** onto a control waveguide **32**. The coupled portion of the light source signal serves as a tapped signal. Although FIG. 1 illustrates a directional coupler operating as the splitter **30**, other signal tapping components can be used as the splitter **30**. Suitable splitters **30** include, but are not limited to, directional couplers, optical couplers, y-junctions, tapered couplers, and Multi-Mode Interference (MMI) devices.

The control waveguide **32** carries the tapped signal to control component **34**. The control component **34** can be in electrical communication with electronics **36**. All or a portion of the control component **34** can be included in the electronics **36**. During operation, the electronics can employ output from the control component **34** to control a process variable of one, two, three, four, or five controlled light signals selected from the group consisting of the tapped signal, the system output signal, the outgoing LIDAR signal, the system output signal, and the LIDAR output signal. Examples of the suitable process variables include the frequency of the controlled light signal and/or the phase of the controlled light signal. The electronics **36** can use output from the control component **34** in a feedback control loop to control the process variable.

The illustrated LIDAR chip includes an optical amplifier **46** that can be operated by the electronics **36**. The optical amplifier includes an amplifier waveguide **48**. The light

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source signal passes through the amplifier waveguide and is amplified before being received at the utility waveguide. The optical amplifier **46** is optional. As a result, the light source signal can be received directly from the light source without being amplified by the amplifier.

FIG. **2** is a schematic of a suitable light source **10** for use with the LIDAR chip of FIG. **1**. The light source **10** includes an optical signal source **50** configured to output a source output signal. A suitable optical signal source **50** includes, but is not limited to a laser source. In some instances, the optical signal source **50** linearly polarized. As a result, the light in the source output signal can include primarily, consist of, and/or consist essentially of light in the first polarization state or light in the second polarization state. In FIG. **1**, the source output signal is illustrated as consisting of and/or consisting essentially of light in the first polarization state (labeled p_1).

The optical signal source **50** includes one or more isolators **52** configured to receive the source output signal and to output the source output signal. The one or more optical isolators can be configured to stop or substantially stop back-reflected light from reaching the optical signal source **50**. Since amplification of light signals processed by the LIDAR system can be a source of back-reflection, the one or more isolators **52** can allow the LIDAR system to have the benefits of amplification without loss in performance of the LIDAR system.

The light source **10** includes a polarization rotator **54** configured to receive the source output signal from the one or more isolators **52** and to output the light source signal. The polarization rotator **54** can be a reciprocal or non-reciprocal polarization rotator. Suitable polarization rotators **54** include, but are not limited to, half-wave plates, rotation of polarization-maintaining fibers, Faraday rotators, MEMS-based polarization rotators and integrated optical polarization rotators using asymmetric y-branches, Mach-Zehnder interferometers and multi-mode interference couplers. When the light in the source output signal is in the first polarization state, the polarization rotator can introduce the second polarization state into the light source signal output from the polarization rotator. Alternately, when the light in the source output signal is in the second polarization state, the polarization rotator can introduce the first polarization state into the light source signal output from the polarization rotator.

Changing the rotation angle of a polarization rotator **54** such as a half-wave plate relative to the source output signal, changes relative amounts of the first polarization state and the second polarization state in the light source signal output. As a result, the rotation angle of the polarization rotator relative to the source output signal can be selected to provide the light source signal output with the desired ratio for the power of light in the first polarization state: the power light in the second polarization state. Suitable ratios for power of light in the first polarization state: power of light in the second polarization state include, but are not limited to, ratios greater than 20:1, 10:1, or 5:1, and/or less than 4:1, 3:1, or 2:1. In some instances, the optical signal source **50** outputs a source output signal that consists of and/or consists essentially of light in the second polarization state and the ratios for power of light in the first polarization state: power of light the second polarization state include, but are not limited to, ratios greater than 20:1, 10:1, or 5:1, and/or less than 4:1, 3:1, or 2:1. In some instances, the optical signal source **50** outputs a source output signal that consists of and/or consists essentially of light in the first polarization state and the ratios for power of light in the first polarization state: power of light the second polarization state include,

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but are not limited to, ratios greater than 20:1, 10:1, or 5:1, and/or less than 4:1, 3:1, or 2:1.

The light source **10** can optionally include one or more beam-shaping components configured to provide an optical signal with the desired shape. For instance, the light source **10** in FIG. **2** includes a first beam-shaping component **56** that receives the source output signal from the light source **10**. The first beam-shaping component **56** can be configured to provide the source output signal with the desired shape. For instance, the first beam-shaping component **56** can be configured to collimate the source output signal such that the source output signal received by the isolator **56** is collimated.

The light source **10** in FIG. **2** includes a second beam-shaping component **58** that receives the light source signal from the polarization rotator **54**. The second beam-shaping component **58** can be configured to provide the source output signal with the desired shape. For instance, the second beam-shaping component **58** can be configured to focus the source output signal at a desired location. In one example, the second beam-shaping component **58** is configured to focus the source output signal such that the source output signal is coupled to the input waveguide **11** or the amplifier waveguide **48**. For instance, the second beam-shaping component **58** is configured to focus the source output signal at a facet of the input waveguide **11**, or at a facet of the amplifier waveguide **48**.

When a LIDAR system includes a light source constructed according to FIG. **2**, the signal output from the polarization rotator **54** can serve as the light source signal output from the light source **10** and received by the LIDAR chip. As a result, the comparative signal and the reference signal can include primarily, consist of, and/or consists essentially of light from the source output signal. As a result, the source output signal can serve as a common source signal for the reference signal and the comparative signal.

All or a portion of the components of the light source such as the light source **50**, the isolator **52**, the polarization rotator **54**, and the beam-shaping components can be positioned on a base **60**. Suitable bases **60** include, but are not limited to, substrates, platforms, desktops and plates. The components can be discrete components that are attached to the base **60**. Suitable mechanisms for attaching discrete components to the base **60** include, but are not limited to, epoxy, solder, and mechanical clamping.

As noted above, the LIDAR chip include one or more waveguides that constrains the optical path of one or more light signals. While the light source can optionally include waveguides, the optical path that the signals travel between components on the light source **10** and/or between the LIDAR chip and a component on the light source **10** can be free space. For instance, the signals can travel through the atmosphere in which the LIDAR chip, the light source **10** and/or a LIDAR system that includes the light source **10** is positioned when traveling between the different components on the light source **10**.

FIG. **3A** illustrates a LIDAR system that includes the LIDAR chip of FIG. **1** and the light source of FIG. **2**. The LIDAR system can include one or more system components that are at least partially located off the LIDAR chip. Examples of suitable system components include, but are not limited to, optical links, beam-shaping components, polarization state rotators, beam steering components, optical splitters, optical amplifiers, and optical attenuators. For instance, the LIDAR system of FIG. **3A** can include one or more beam-shaping components **130** that receive the LIDAR output signal from the LIDAR chip and outputs a

shaped signal. The one or more beam-shaping components **130** can be configured to provide the shaped signal with the desired shape. For instance, the one or more beam-shaping components **130** can be configured to output a shaped signal that is focused, diverging or collimated. In FIG. 3A, the one or more beam-shaping components **130** is a lens that is configured to output a collimated shaped signal.

The LIDAR systems of FIG. 3A includes one or more polarization rotators **132** that receive the shaped signal and outputs a rotated signal. In some instances, the one or more polarization rotators **132** are configured to rotate the polarization state of the shaped signal by $n \cdot 90^\circ + 45^\circ$ where n is 0 or an even integer. Suitable polarization rotators **132** include, but are not limited to, non-reciprocal polarization rotators such as Faraday rotators.

The LIDAR system of FIG. 3A can optionally include one or more beam steering components **134** that receive the rotated signal from the one or more polarization rotators **132** and that output the system output signal. For instance, **3** illustrates a beam steering component **134** that receive the rotated signal from a polarization rotator **132**. The electronics **36** can operate the one or more beam steering component **134** so as to steer the system output signal to different sample regions **135**. The sample regions can extend away from the LIDAR system to a maximum distance for which the LIDAR system is configured to provide reliable LIDAR data. The sample regions can be stitched together to define the field of view. For instance, the field of view of for the LIDAR system includes or consists of the space occupied by the combination of the sample regions.

Suitable beam steering components include, but are not limited to, movable mirrors, MEMS mirrors, optical phased arrays (OPAs), optical gratings, actuated optical gratings and actuators that move the LIDAR chip.

When the system output signal is reflected by an object **136** located outside of the LIDAR system and the LIDAR, at least a portion of the reflected light returns to the LIDAR system as a system return signal. While the reflection by the object can changes the polarization state of a portion of the light in the system output signal, the polarization state is generally retained. As a result, the light in the system return signal is primarily in the same polarization state as the light in the system output signal.

When the LIDAR system includes one or more beam steering components **134**, the one or more beam steering components **134** can receive at least a portion of the system return signal from the object **136**. The one or more polarization rotators **132** can receive at least a portion of the system return signal from the object **136** or from the one or more beam steering components **134**. The one or more polarization rotators **132** can output a rotated return signal. The one or more beam-shaping components **130** receive the rotated return signal from the one or more polarization rotators **132** and output the LIDAR input signal that is received by the LIDAR chip.

When the one or more polarization rotators **132** are configured to rotate the polarization state of the shaped signal by $n \cdot 90^\circ + 45^\circ$ where n is 0 or an even integer, when the one or more polarization rotators **132** are non-reciprocal, the one or more polarization rotators **132** also rotate the polarization state of the system return signal such that the polarization state of the rotated return signal is rotated by $n \cdot 90^\circ - 45^\circ$ relative to the polarization state of the system return signal. As a result, the polarization state of the rotated return signal is changed by $n \cdot 180^\circ + 90^\circ$ (where n is 0 or an even integer) relative to the polarization state of the shaped signal. Accordingly, the polarization state of the LIDAR

input signal is increased by $n \cdot 180^\circ + 90^\circ$ (where n is 0 or an even integer) relative to the polarization state of the LIDAR output signal. For instance, when the LIDAR output signal includes primarily, consists essentially of, and/or consists of light in the first polarization state, the LIDAR input signal includes primarily, consists essentially of, and/or consists of light in the second polarization state and in some instances, can exclude or substantially exclude light in the first polarization state. Alternately, when the LIDAR output signal includes primarily, consists essentially of, and/or consists of light in the second polarization state, the LIDAR input signal includes primarily, consists essentially of, or consists of light in the first polarization state.

The LIDAR system of FIG. 3A include an optional optical link **138** that carries optical signals to the one or more system components from the LIDAR chip. For instance, the LIDAR system of FIG. 3A includes an optical fiber or a polarization maintaining optical fiber configured to carry the LIDAR output signal to the beam-shaping component **130**. The use of the optical link **138** allows the source of the system output signal to be located remote from the LIDAR chip. Although the illustrated optical link **138** is an optical fiber, other optical links **138** can be used. Suitable optical links **138** include, but are not limited to, free space optical links and waveguides. When the LIDAR system excludes an optical link, the one or more beam-shaping components **130** can receive the assembly output signal directly from the LIDAR chip.

Although the LIDAR system of FIG. 1 through FIG. 3A illustrates the splitter as a 2x2 splitter, the splitter can be a 1x2 splitter such as a 1x2 polarization splitter. For instance, the splitter **14** need not output the reference signal on the reference waveguide **28**. Accordingly, a reference waveguide **28** need not terminate at the splitter **14**. For instance, the light source **10** can include a splitter that taps off a portion of the source output signal to serve as a reference precursor signal. Rather than the polarization rotator **54** receiving the source output signal, the polarization rotator **54** can receive the reference precursor signal and be configured to change the polarization state of the reference precursor signal from the first polarization state to the second polarization state or from the second polarization state to the first polarization state. Light from the output of the polarization rotator **54** can serve as a reference signal that is received by a reference waveguide on the LIDAR chip. The reference waveguide can carry the received reference signal to the processing component **26**. The presence of the polarization rotator **54** in the light source **10** provides the reference signals and the comparative signal received by the processing component **26** with the same polarization state.

FIG. 3B illustrates the LIDAR system of FIG. 3A modified to include a 1x2 polarization splitter in place of the 2x2 splitter illustrated in the LIDAR system of FIG. 3A. The light source **10** is modified to include a beam splitter **140** that receives the source output signal from the one or more isolators **52**. The beam splitter splits the light source signal into a first portion of the light source signal and a second portion of the reference precursor signal. The first portion of the light source signal serves as the light source signal that is coupled to the input waveguide **11** or the amplifier waveguide **48** on the LIDAR chip. The second portion of the light source signal serves as a reference precursor signal.

The polarization rotator **54** receives the reference precursor signal and outputs a second precursor signal. The polarization rotator **54** is configured to change the polarization state of the reference precursor signal. As a result, when the

reference precursor signal has the first polarization state, the second precursor signal has the second polarization state as illustrated in FIG. 3B. Alternately, when the reference precursor signal has the second polarization state, the second precursor signal has the first polarization state.

The second precursor signal is received by a reference waveguide 28 on the LIDAR chip. For instance, the reference waveguide 28 can terminate at a facet 144 and the second precursor signal can enter the reference waveguide 28 through the facet 144. The portion of the second precursor signal that enters the reference waveguide 28 can serve as the reference signal that the reference waveguide 28 carries to the processing component 26 for further processing. As is evident from FIG. 3B, the reference signal and the comparative signal received by the processing component 26 are in the same polarization state.

The light source 10 of FIG. 3B includes a third beam-shaping component 146 that receives the second precursor source signal from the polarization rotator 54. The third beam-shaping component 146 can be configured to provide the source output signal with the desired shape. For instance, the third beam-shaping component 146 can be configured to focus the second precursor signal at a desired location. In one example, the third beam-shaping component 146 is configured to focus the second precursor signal such that the second precursor signal is coupled to the reference waveguide 28. For instance, the third beam-shaping component 146 is configured to focus the second precursor signal at the facet 144 of the reference waveguide 28.

The light source 10 can also include one or more direction changing components such as mirrors or prisms. FIG. 3B illustrates the light source 10 including a mirror 148 as a direction-changing component that redirects the reference precursor signal from the splitter 140 to the polarization rotator 54.

FIG. 4A through FIG. 4B illustrate an example of a processing component that is suitable for use as the processing component 26 in the above LIDAR systems. The processing component 26 receives the comparative signal from the comparative waveguide 24 disclosed in the context of FIG. 1. The comparative waveguide 24 carries the comparative signal contribution to a light-combining component 154. The processing component 26 receives the reference signal contribution from the reference waveguide 28 disclosed in the context of FIG. 1. The reference waveguide 28 carries the reference signal contribution to the light-combining component 154. The light-combining component 154 combines the comparative signal and the reference signal into a composite signal. As noted above, the reference signal and the comparative signal received by the processing component 26 each includes primarily, consists essentially of, and/or consists of light in the same polarization state. As a result, the light-combining component 154 combines signals of the same polarization state. Due to the difference in frequencies between the comparative signal and the reference signal, the composite signal is beating between the comparative signal and the reference signal.

The light-combining component 154 also splits the resulting composite signal onto a first detector waveguide 156 and a second detector waveguide 158. The first auxiliary detector waveguide 156 carries a first portion of the composite signal to a first light sensor 160 that converts the first portion of the composite signal to a first electrical signal. The second detector waveguide 158 carries a second portion of the composite signal to a second light sensor 162 that converts the second portion of the composite signal to a second

electrical signal. Examples of suitable light sensors include germanium photodiodes (PDs), and avalanche photodiodes (APDs).

In some instances, the light-combining component 154 splits the composite signal such that the comparative signal contribution in the included in the first portion of the composite signal is phase shifted by 180° relative to comparative signal contribution in the second portion of the composite signal but the reference signal contribution in the second portion of the composite signal is in-phase with the reference signal contribution in the first portion of the composite signal. Alternately, the light-combining component 154 splits the composite signal such that the reference signal contribution in the first portion of the composite signal is phase shifted by 180° relative to the reference signal contribution in the second portion of the composite signal but the comparative signal contribution in the first portion of the composite signal is in-phase with the portion of the comparative signal in the second portion of the composite signal. Examples of suitable light sensors include germanium photodiodes (PDs), and avalanche photodiodes (APDs).

FIG. 4B provides a schematic of the relationship between the electronics 36 and the light sensors in a processing component 138. The symbol for a photodiode is used to represent the first light sensor 160 and the second light sensor 162, but one or more of these sensors can have other constructions. In some instances, all of the components illustrated in the schematic of FIG. 4B are included on the LIDAR chip. In some instances, the components illustrated in the schematic of FIG. 4B are distributed between the LIDAR chip and electronics located off of the LIDAR chip.

The electronics 36 can connect the first light sensors 160 and the second light sensors 162 as a balanced detector 164. For instance, the electronics can connect the first light sensors 160 in series with the second light sensor 162 as illustrated in FIG. 4B. The serial connection between the first light sensor 160 and the second light sensors 162 carries the output from the balanced detector as a data signal. The data signal can be carried on a sensor output line 166 and can serve as an electrical representation of the composite signals.

The electronics 36 include a transform mechanism 168 configured to perform a mathematical transform on the data signal. The transform mechanism 168 includes an Analog-to-Digital Converter (ADC) 170 that receives the data signal from the sensor output line 166. The Analog-to-Digital Converter (ADC) 170 converts the data signal from an analog form to a digital form and outputs a digital data signal. The digital data signal is a digital representation of the data signal.

The transform mechanism 168 includes a mathematical operation component 172 configured to receive the digital data signal. The mathematical operation component 172 is configured to perform a mathematical operation on the received digital data signal. Examples of suitable mathematical operations include, but are not limited to, mathematical transforms such as Fourier transforms. In one example, the mathematical operation component 172 performs a Fourier transform on the digital signal so as to convert from the time domain to the frequency domain. The mathematical transform can be a real transform such as a real Fast Fourier Transform (FFT). A real Fast Fourier Transform (FFT) can provide an output that indicates magnitude as a function of frequency. As a result, a peak in the output of the Fast Fourier Transform can occur at and/or indicate the correct solution for the beat frequency of the beating signals. The mathematical operation component 172

can execute the attributed functions using firmware, hardware or software or a combination thereof.

The electronics include a LIDAR data generator **174** that receives the output from the transform component **168**. The LIDAR data generator **174** can perform a peak find on the output of the transform component **168** to identify the peak in the frequency of the output of the transform component **168**. The LIDAR data generator **174** treats the frequency at the identified peak as the beat frequency of the beating signals that each results from all or a portion of a comparative signal beating against all or a portion of a reference signal. The LIDAR data generator **174** can use the identified beat frequencies in combination with the frequency pattern of the LIDAR output signal and/or the system output signal to generate LIDAR data that indicates the radial velocity and/or distance between the LIDAR system and one or more objects located in the sample region.

As shown in FIG. 4B, the sensor output line **166** that carries the data signals can optionally include an amplifier **176**. Suitable amplifiers **176** include, but are not limited to, transimpedance amplifiers (TIAs).

FIG. 4C has a solid line that shows an example of a suitable frequency pattern for the LIDAR output signal and accordingly the system output signal. Accordingly, the solid line also represents the frequency pattern for the reference signal. FIG. 4C shows the frequency versus time pattern over a sequence of two cycles, labeled cycle_j and cycle_{j+1}. In some instances, the frequency versus time pattern is repeated in each cycle as shown in FIG. 4C. The illustrated cycles do not include re-location periods and/or re-location periods are not located between cycles. As a result, FIG. 4C illustrates the results for a continuous scan of the system output signal to multiple different sample regions in a field of view.

Each cycle includes K data periods that are each associated with a period index k and are labeled DP_k. In the example of FIG. 4C, each cycle includes two data periods (with k=1 and 2). In some instances, the frequency versus time pattern is the same for the data periods that correspond to each other in different cycles as is shown in FIG. 4C. Corresponding data periods are data periods with the same period index. As a result, each data period DP₁ can be considered corresponding data periods for that same channel index (i) and the associated frequency versus time patterns are the same in FIG. 4C. At the end of a cycle, the electronics return the frequency to the same frequency level at which it started the previous cycle.

During each data period, the frequency of the system output signal is varied at a constant rate. The rate can be zero but at least a portion of the data periods in each cycle have the system output signal varied at a non-zero rate. The direction and/or rate of the frequency change changes at the change of data periods from the same cycle. For instance, during the data period DP₁ and the data period DP₂, the electronics operate the light source such that the frequency of the system output signal changes at a linear rate a. The direction of the frequency change during the data period DP₁ is the opposite of the direction of the frequency change during the data period DP₂.

The beat frequencies (f_{LDP}) from two or more different data periods in the same cycle can be combined to generate the LIDAR data. For instance, the beat frequency determined from DP₁ in FIG. 4C can be combined with the beat frequency determined from DP₂ in FIG. 4C to determine the LIDAR data for a sample region. As an example, the following equation applies during a data period where electronics increase the frequency of the outgoing LIDAR signal during the data period such as occurs in data period

DP₁ of FIG. 4C: $f_{ub} = -f_d + \alpha\tau$ where f_{ub} is the beat frequency determined from the output of the mathematical operation component **172**, f_d represents the Doppler shift ($f_d = 2v f_c / c$) where f_c represents the optical frequency (f_c), c represents the speed of light, v is the radial velocity between the reflecting object and the LIDAR system where the direction from the reflecting object toward the LIDAR system is assumed to be the positive direction, and c is the speed of light. The following equation applies during a data period where electronics decrease the frequency of the outgoing LIDAR signal such as occurs in data period DP₂ of FIG. 4C: $f_{db} = -f_d - \alpha\tau$ where f_d is the beat frequency determined from the output of the mathematical operation component **172**. In these two equations, f_d and τ are unknowns. The electronics solve these two equations for the two unknowns. The radial velocity for the sample region then be determined from the Doppler shift ($v = c * f_d / (2f_c)$) and/or the separation distance for that sample region can be determined from $c * \tau / 2$. Since the LIDAR data can be generated for each corresponding frequency pair output by the transform, separate LIDAR data can be generated for each of the objects in a sample region. Accordingly, the electronics can determine more than one radial velocity and/or more than one radial separation distance from a single sampling of a single sample region in the field of view.

Suitable platforms for the LIDAR chips include, but are not limited to, silica, indium phosphide, and silicon-on-insulator wafers. FIG. 5 is a cross-section of a portion of a chip constructed from a silicon-on-insulator wafer. A silicon-on-insulator (SOI) wafer includes a buried layer **310** between a substrate **312** and a light-transmitting medium **314**. In a silicon-on-insulator wafer, the buried layer **310** is silica while the substrate **312** and the light-transmitting medium **314** are silicon. The substrate **312** of an optical platform such as an SOI wafer can serve as the base for the entire LIDAR chip. For instance, the optical components shown on the LIDAR chips of FIG. 1 through FIG. 2 can be positioned on or over the top and/or lateral sides of the substrate **312**.

FIG. 5 is a cross section of a portion of a LIDAR chip that includes a waveguide construction that is suitable for use in LIDAR chips constructed from silicon-on-insulator wafers. A ridge **316** of the light-transmitting medium extends away from slab regions **318** of the light-transmitting medium. The light signals are constrained between the top of the ridge **316** and the buried oxide layer **310**.

The dimensions of the ridge waveguide are labeled in FIG. 5. For instance, the ridge has a width labeled w and a height labeled h. A thickness of the slab regions is labeled T. For LIDAR applications, these dimensions can be more important than other dimensions because of the need to use higher levels of optical power than are used in other applications. The ridge width (labeled w) is greater than 1 μm and less than 4 μm, the ridge height (labeled h) is greater than 1 μm and less than 4 μm, the slab region thickness is greater than 0.5 μm and less than 3 μm. These dimensions can apply to straight or substantially straight portions of the waveguide, curved portions of the waveguide and tapered portions of the waveguide(s). Accordingly, these portions of the waveguide will be single mode. However, in some instances, these dimensions apply to straight or substantially straight portions of a waveguide. Additionally or alternately, curved portions of a waveguide can have a reduced slab thickness in order to reduce optical loss in the curved portions of the waveguide. For instance, a curved portion of a waveguide can have a ridge that extends away from a slab region with a thickness greater than or equal to 0.0 μm and less than 0.5

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μm. While the above dimensions will generally provide the straight or substantially straight portions of a waveguide with a single-mode construction, they can result in the tapered section(s) and/or curved section(s) that are multi-mode. Coupling between the multi-mode geometry to the single mode geometry can be done using tapers that do not substantially excite the higher order modes. Accordingly, the waveguides can be constructed such that the signals carried in the waveguides are carried in a single mode even when carried in waveguide sections having multi-mode dimensions. The waveguide construction disclosed in the context of FIG. 5 is suitable for all or a portion of the waveguides on LIDAR chips constructed according to FIG. 1 through FIG. 2.

Light sensors that are interfaced with waveguides on a LIDAR chip can be a component that is separate from the chip and then attached to the chip. For instance, the light sensor can be a photodiode, or an avalanche photodiode. Examples of suitable light sensor components include, but are not limited to, InGaAs PIN photodiodes manufactured by Hamamatsu located in Hamamatsu City, Japan, or an InGaAs APD (Avalanche Photo Diode) manufactured by Hamamatsu located in Hamamatsu City, Japan. These light sensors can be centrally located on the LIDAR chip. Alternately, all or a portion the waveguides that terminate at a light sensor can terminate at a facet located at an edge of the chip and the light sensor can be attached to the edge of the chip over the facet such that the light sensor receives light that passes through the facet. The use of light sensors that are a separate component from the chip is suitable for all or a portion of the light sensors selected from the group consisting of the first light sensor and the second light sensor.

As an alternative to a light sensor that is a separate component, all or a portion of the light sensors can be integrated with the chip. For instance, examples of light sensors that are interfaced with ridge waveguides on a chip constructed from a silicon-on-insulator wafer can be found in Optics Express Vol. 15, No. 21, 13965-13971 (2007); U.S. Pat. No. 8,093,080, issued on Jan. 10 2012; U.S. Pat. No. 8,242,432, issued Aug. 14 2012; and U.S. Pat. No. 6,108,847, issued on Aug. 22, 2000 each of which is incorporated herein in its entirety. The use of light sensors that are integrated with the chip are suitable for all or a portion of the light sensors selected from the group consisting of the first light sensor and the second light sensor.

When the LIDAR chip includes one or more amplifiers, one or more amplifiers can be integrated onto the platform of the LIDAR chip. For instance, one or more amplifiers can be integrated onto a LIDAR chip constructed on a silicon-on-insulator wafer. An example of an amplifier construction that can be integrated onto a silicon-on-insulator wafer can be found in U.S. patent application Ser. No. 13/317,340, filed on Oct. 14 2011, entitled Gain Medium Providing Laser and Amplifier Functionality to Optical Devices, and incorporated herein in its entirety.

FIG. 6A is a perspective view of a portion of a LIDAR chip that includes an interface for optically coupling the LIDAR chip with an amplifier chip. The illustrated portion of the LIDAR chip includes a stop recess 330 sized to receive the amplifier. The stop recess 330 extends through the light-transmitting medium 324 and into the base 321. In the illustrated version, the stop recess 330 extends through the light-transmitting medium 324, the buried layer 320, and into the substrate 322.

A facet 342 of the light-transmitting medium 324 serves as a lateral side of the stop recess 30. The facet 342 can be a facet of a waveguide 344 depending on the application of

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the amplifier. For instance, the facet 342 can be a facet of a source waveguide when the amplifier is used as disclosed in the context of FIG. 1 or a facet of a utility waveguide when the amplifier is used as disclosed in the context of FIG. 8. Although not shown, the facet 342 can include an anti-reflective coating. A suitable anti-reflective coating includes, but is not limited to, single-layer coatings such as silicon nitride or aluminum oxide, or multi-layer coatings, which may contain silicon nitride, aluminum oxide, and/or silica.

One or more stops 332 extend upward from a bottom of the stop recess 330. For instance, FIG. 6A illustrates four stops 332 extending upward from the bottom of the stop recess 330. The stops 332 include a cladding 334 positioned on a base portion 336. The substrate 322 can serve as the base portion 336 of the stops 332 and the stop 332 can exclude the buried layer 320. The portion of the substrate 322 included in the stops 332 can extend from the bottom of the stop recess 330 up to the level of the buried layer 320. For instance, the stops 332 can be formed by etching through the buried layer 320 and using the underlying substrate 322 as an etch-stop. As a result, the location of the top of the base portion 336 relative to the optical mode of a light signal in the waveguide 384 is well known because the buried layer 320 defines the bottom of the second waveguide and the top of the base portion 336 is located immediately below the buried layer 320. The cladding 334 can be formed on base portion 336 of the stops 332 so as to provide the stops 332 with a height that will provide the desired alignment between the waveguide 384 and an amplifier waveguide on an amplifier chip.

Attachment pads 338 are positioned on the bottom of the stop recess 330. The attachment pads 338 can be used to immobilize the amplifier chip relative to the LIDAR chip once the amplifier chip is positioned on the LIDAR chip. In some instances, the attachment pads 338 also provide electrical communication between the LIDAR chip and one or more amplifiers on an amplifier chip. Suitable attachment pads 338 include, but are not limited to, solder pads.

FIG. 6B is a perspective view of one embodiment of an amplifier chip. The illustrated amplifier chip is within the class of devices known as planar optical devices. The amplifier chip includes an amplifier waveguide 346 defined in a gain medium 340. Suitable gain media include, but are not limited to, InP, InGaAsP, and GaAs.

Trenches 374 extending into the gain medium 340 define a ridge 376 in the gain medium 340. The ridge 376 defines the amplifier waveguide 346. In some instances, the gain medium 340 includes one or more layers 341 in the ridge and/or extending across the ridge 376. The one or more layers 341 can be positioned between different regions of the gain medium 340. The region of the gain medium 340 above the one or more layers 341 can be the same as or different from the region of the gain medium 340 below the one or more layers 341. The layers can be selected to constrain light signals guided through the amplifier waveguide 346 to a particular location relative to the ridge 376. Each of the layers 341 can have a different composition of a material that includes or consists of two or more components of selected from a group consisting of In, P, Ga, and As. In one example, the gain medium 340 is InP and the one or more layers 341 each includes Ga and As in different ratios.

The amplifier waveguide 346 provides an optical pathway between a first facet 350 and the second facet 352. Although not shown, the first facet 350 and/or the second facet 352 can optionally include an anti-reflective coating. A suitable anti-reflective coating includes, but is not limited to, single-

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layer coatings such as silicon nitride or aluminum oxide, or multi-layer coatings that may contain silicon nitride, aluminum oxide, and/or silica.

The amplifier chip includes one or more attachment pads **354** that can be employed to immobilize the amplifier chip relative to the LIDAR chip. Suitable attachment pads **354** include, but are not limited to, solder pads.

The amplifier chip includes a first conductor **360** on the ridge and a second conductor **362** that is both under the gain medium and under the ridge **376**. The first conductor **360** is in electrical communication with an attachment pad **354**. Suitable methods for providing electrical communication between the first conductor **360** and the attachment pad **354** include, but are not limited to, conducting metal traces.

The amplifier chip also includes one or more alignment recesses **356**. The dashed lines in FIG. 6B show the depth and shape of one of the alignment recesses **356**.

FIG. 6C and FIG. 6D illustrates the LIDAR chip of FIG. 6A interfaced with the amplifier chip of FIG. 6B. FIG. 6C is a topview of the LIDAR system. FIG. 6D is a sideview of a cross section of the system taken through the waveguide **384** on the LIDAR chip and an amplifier waveguide **346** on the amplifier chip. For instance, the cross section of FIG. 6D can be taken a long a line extending through the brackets labeled B in FIG. 6C. FIG. 6C and FIG. 6D each includes dashed lines that illustrate features that are located behind other features in the system. For instance, FIG. 6C includes dashed lines showing the ridge **376** of the amplifier waveguide **346** even though the ridge **376** is located under the gain medium **340**. Additionally, FIG. 6D includes dashed lines that illustrate the locations of the portion of the stops **332** and alignment recesses **356** located behind the ridge **376** of the amplifier waveguide **346**. FIG. 6D also includes dashed lines that illustrate the location where the ridge **326** of waveguide **384** interfaces with the slab regions **328** that define the waveguide **384** also dashed lines that illustrate the location where the ridge **376** of the amplifier waveguide **346** interfaces with slab regions **374** of the amplifier chip.

The amplifier chip is positioned in the stop recess **330** on the LIDAR chip. The amplifier chip is positioned such that the ridge **376** of the amplifier waveguide **346** is located between the bottom of the amplifier chip and the base **321** of the LIDAR chip. Accordingly, the amplifier chip is inverted in the stop recess **330**. Solder or other adhesive **358** contacts the attachment pads **338** on the bottom of the stop recess **330** and the attachment pads **354** on the amplifier chip. For instance, the solder or other adhesive **358** extends from an attachment pad **338** on the bottom of the stop recess **330** to an attachment pad **354** on the auxiliary device. Accordingly, the solder or other adhesive **358** immobilizes the auxiliary device relative to the LIDAR chip.

The facet **342** of the waveguide **384** is aligned with the first facet **350** of the amplifier waveguide **346** such that the waveguide **384** and the amplifier waveguide **346** can exchange light signals. As shown by the line labeled A, the system provides a horizontal transition path in that the direction that light signals travel between the LIDAR chip and the amplifier chip is parallel or is substantially parallel relative to an upper and/or lower surface of the base **321**. A top of the first facet **350** of the amplifier waveguide **346** is at a level that is below the top of the facet **342** of the utility waveguide.

The one or more stops **332** on the LIDAR chip are each received within one of the alignment recesses **356** on the auxiliary device. The top of each stop **332** contacts the bottom of the alignment recess **356**. As a result, the interaction between stops **332** and the bottom of the alignment

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recesses **356** prevent additional movement of the amplifier chip toward the LIDAR chip. In some instances, the auxiliary device rests on top of the stops **332**.

As is evident from FIG. 6D, the first facet **350** of the amplifier waveguide **346** is vertically aligned with the facet **342** of the waveguide **384** on the LIDAR chip. As is evident from FIG. 6C, the first facet **350** of the amplifier waveguide **346** is horizontally aligned with the facet **342** of the waveguide **384** on the LIDAR chip. The horizontal alignment can be achieved by alignment of marks and/or features on the amplifier chip and the LIDAR chip.

The vertical alignment can be achieved by controlling the height of the stops **332** on the LIDAR chip. For instance, the cladding **334** on the base portion **336** of the stops **332** can be grown to the height that places the first facet **350** of the amplifier waveguide **346** at a particular height relative to the facet **342** of the waveguide **384** on the LIDAR chip. The desired cladding **334** thickness can be accurately achieved by using deposition techniques such as evaporation, plasma enhanced chemical vapor deposition (PECVD), and/or sputtering to deposit the one or more cladding layers. As a result, one or more cladding layers can be deposited on the base portion **336** of the stops **332** so as to form the stops **332** to a height that provides the desired vertical alignment. Suitable materials for layers of the cladding **334** include, but are not limited to, silica, silicon nitride, and polymers.

In FIG. 6D, the first facet **350** is spaced apart from the facet **342** by a distance labeled D. Since the amplifier waveguide is optically aligned with only one waveguide, the first facet **350** can be closer to the facet **342** than was possible with prior configurations. For instance, the distance between the first facet **350** and the facet **342** can be less than 5 μm , 3 μm , or 1 μm and/or greater than 0.0 μm . In FIG. 1D, the atmosphere in which the LIDAR chip is positioned is located in the gap between the first facet **350** and the facet **342**; however, other gap materials can be positioned in the gap. For instance, a solid gap material can be positioned in the gap. Examples of suitable gap materials include, but are not limited to, epoxies and polymers.

The LIDAR chip includes electrical pathways **380** on the light-transmitting medium **324**. The electrical pathways **380** can optically include contact pads and can be in electrical communication with the electronics. Although not illustrated, one of the electrical pathways **380** can be in electrical communication with the contact pad **354**. Since the contact pad **354** is in electrical communication with the first conductor **360**, the contact pad **354** provides electrical communication between the first conductor **360** and the electronics. Another one of the electrical pathways **380** can be in electrical communication with the second conductor **362**. Suitable methods for providing electrical communication between the second conductor **362** and the electrical pathway **380** include, but are not limited to, wire bonding. Suitable electrical pathways **380** include, but are not limited to, metal traces.

The electronics can use the electrical pathways **380** to apply electrical energy to the portion of the amplifier between the first conductor **360** and the second conductor **362**. The electronics can apply the electrical energy so as to drive an electrical current through the amplifier waveguide **346**. The electrical current through the gain medium provides the amplification of light signals guided in the amplifier waveguide **346**.

The polarization splitter can be a separate component that is attached to the silicon-on-insulator platform or can be integrated with the silicon-on-insulator platform. Examples of polarization splitters that can be used with a silicon-on-

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insulator platform include, but are not limited to, Mach-Zehnder Interferometers, and zero-order Arrayed Waveguide Gratings (AWGs). FIG. 7 is a topview of a Mach-Zehnder interferometer that is modified to serve as a polarization splitter 14 that can be integrated with the silicon-on-insulator platform. The splitter includes a first branch waveguide 396 and a second branch waveguide 398 between a first optical coupler 400 and a second optical coupler 402. The input waveguide 11 and the reference waveguide 28 converge at the first optical coupler 400. Additionally, the utility waveguide 12 and comparative waveguide 24 converge at a second optical coupler 402. The input waveguide carries the input signal to the first optical coupler 400. The first optical coupler 400 divides the input signal into a first input signal that is output on the first branch waveguide 396 and a second input signal that is output on the second branch waveguide 398. The second optical coupler 402 combines the first input signal and the second input signal. The first branch waveguide 396 and the second branch waveguide 398 can have different lengths such that there is a phase differential between the first input signal and the second input signal at the second optical coupler 402. Additionally, the first branch waveguide 396 and the second branch waveguide 398 have different birefringence levels. As a result of the phase differential and the birefringence differential, the second optical coupler 402 outputs the portion of the combined signal that is in the first polarization state is output on the utility waveguide 12 and the portion of the combined signal that is in the second polarization state is output on the comparative waveguide 24. Additionally, the second optical coupler 402 divides the incoming LIDAR signal into a first incoming LIDAR signal that is output on the first branch waveguide 396 and a second incoming LIDAR signal that is output on the second branch waveguide 398. The first optical coupler 400 combines the incoming LIDAR signal and the second incoming LIDAR signal. As a result of the phase differential and the birefringence differential, the first optical coupler 400 outputs the portion of the combined signal that is in the second polarization state on the reference waveguide 28. Additionally, the first optical coupler 400 outputs any of the combined signal that is in the first polarization state on the input waveguide 11.

One way to provide the birefringence differential is to provide the first branch waveguide 396 and the second branch waveguide 398 with different cross sectional dimensions for all or a portion of their length where the cross section is taken transverse or perpendicular to the longitudinal axis of the waveguide. For instance, FIG. 7 shows a portion of the first branch waveguide 396 as having a smaller width than the corresponding portion of the second branch waveguide 398. The reduced width provides the first branch waveguide 396 with a different birefringence than the second branch waveguide 398. In some instances, the first branch waveguide 396 and the second branch waveguide 398 are constructed such that the first branch waveguide 396 has a birefringence that is within the birefringence of the second branch waveguide 398 $\pm 1\%$, $\pm 5\%$, or $\pm 10\%$.

The first branch waveguide 396 and the second branch waveguide 398 can be constructed as disclosed in the context of FIG. 5.

Suitable electronics can include, but are not limited to, a controller that includes or consists of analog electrical circuits, digital electrical circuits, processors, microprocessors, digital signal processors (DSPs), Field Programmable Gate Arrays (FPGAs), computers, microcomputers, or combinations suitable for performing the operation, monitoring

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and control functions described above. In some instances, the controller has access to a memory that includes instructions to be executed by the controller during performance of the operation, control and monitoring functions. Although the electronics are illustrated as a single component in a single location, the electronics can include multiple different components that are independent of one another and/or placed in different locations. Additionally, as noted above, all or a portion of the disclosed electronics can be included on the chip including electronics that are integrated with the chip.

Although the light source 10 is illustrated as being located on the LIDAR chip, the light source can be located off the LIDAR chip. For instance, the LIDAR chip can receive the outgoing LIDAR signal from an optical fiber.

Other embodiments, combinations and modifications of this invention will occur readily to those of ordinary skill in the art in view of these teachings. Therefore, this invention is to be limited only by the following claims, which include all such embodiments and modifications when viewed in conjunction with the above specification and accompanying drawings.

The invention claimed is:

1. A system, comprising:

a LIDAR system includes a polarizing-beam splitter configured to receive an input light signal from an input waveguide and output an outgoing LIDAR signal on a utility waveguide,
the outgoing LIDAR signal including light from the input signal,
the LIDAR system being configured to output a system output signal that includes light from the outgoing LIDAR signal, the system output signal can be reflected by an object located outside of the LIDAR system such that a reflected portion of the system output signal serves as a system return signal; and
the polarizing-beam splitter configured to receive an incoming LIDAR signal from the utility waveguide,
the incoming LIDAR signal including light from the system return signal,
the polarizing-beam splitter outputting at least a portion of the incoming LIDAR signal on a comparative waveguide, and
the polarizing-beam splitter being configured to output a first portion of the input signal on the utility waveguide as the outgoing LIDAR signal and to output a second portion of the input signal on a reference waveguide as a reference signal.

2. The system of claim 1, wherein the utility waveguide is configured to guide the outgoing LIDAR signal and the incoming LIDAR signal in opposing directions.

3. The system of claim 1, wherein the outgoing LIDAR signal is primarily in a first polarization state and the incoming LIDAR signal is primarily in a second polarization state, the first polarization state being different from the second polarization state.

4. The system of claim 1, wherein the outgoing LIDAR signal is primarily in a first polarization state and the reference signal is primarily in a second polarization state, the first polarization state being different from the second polarization state.

5. The system of claim 4, wherein the incoming LIDAR signal is primarily in the second polarization state.

6. The system of claim 1, wherein the at least the portion of the incoming LIDAR signal on the comparative waveguide serves as a comparative signal, and the LIDAR chip

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is configured to generate a composite signal having light from the reference signal beating with light from the comparative signal.

7. The system of claim 1, wherein the reference signal consists of light that has not exited from the LIDAR chip.

8. The system of claim 6, wherein the light from the reference signal that is beating with the light from the comparative signal is in the same polarization state as the light from the comparative signal that is beating with the light from the reference signal.

9. The system of claim 8, wherein the comparative signal does not include light from the reference signal.

10. The system of claim 6, further comprising:

electronics configured to use a beat frequency of the composite signal in a determination of LIDAR data that indicates a distance and/or radial velocity between the LIDAR system and the object.

11. The system of claim 1, wherein the LIDAR chip includes a silicon-on-insulator platform.

12. The system of claim 1, wherein the polarizing-beam splitter is a Mach-Zehnder interferometer.

13. The system of claim 12, wherein the interferometer includes multiple branch waveguides that carry light signals between optical couplers and different branch waveguides have different levels of birefringence.

14. A LIDAR system, comprising:

a LIDAR chip with a utility waveguide configured to guide an outgoing LIDAR signal and an incoming LIDAR signal,

the incoming LIDAR signal including light from the LIDAR output signal after an object located outside of the LIDAR system reflects the light from the LIDAR output signal;

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the LIDAR chip including a polarizing-beam splitter configured to couple a portion of the outgoing LIDAR signal from the utility waveguide onto a reference waveguide as a reference signal and to couple a portion of the incoming LIDAR signal from the utility waveguide onto a comparative waveguide as a comparative signal;

the LIDAR chip configured to generate a composite signal having light from the reference signal beating with light from the comparative signal; and

electronics configured to use a beat frequency of the composite signal as a variable in a calculation of LIDAR data that indicates a distance and/or radial velocity between the LIDAR system and the object.

15. The system of claim 14, wherein the utility waveguide is configured to guide the outgoing LIDAR signal and the incoming LIDAR signal in opposing directions.

16. The system of claim 14, wherein the outgoing LIDAR signal is primarily in a first polarization state and the incoming LIDAR signal is primarily in a second polarization state, the first polarization state being different from the second polarization state.

17. The system of claim 14, wherein the outgoing LIDAR signal is primarily in a first polarization state and the reference signal is primarily in a second polarization state, the first polarization state being different from the second polarization state.

18. The system of claim 14, wherein the light from the reference signal that is beating with the light from the comparative signal is in the same polarization state as the light that is from the comparative signal and is beating with the light from the reference signal.

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